

Gesture Controlled LED System using

Computer Vision

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1. Introduction-

This project implements a vision-based hand gesture controller that allows LEDs to be controlled using finger movements.

The system uses:

- OpenCV for webcam access and image processing
- MediaPipe for hand landmark detection
- Python for gesture processing logic
- Arduino Uno for hardware interfacing
- LEDs for visual output

2. Objectives-

The main objectives of the project are:

1. To detect hand gestures in real time using a webcam.
2. To identify which fingers are open or closed.

3. To send commands from Python to Arduino using serial communication.
4. To control LEDs based on detected gestures.
5. To create a low-cost real-time gesture control system.

3. Components Required-

Component	Quantity
Arduino Uno	1
LEDs	5
220 ohm resistors	5
Breadboard	1
Jumper wires	multiple
USB cable	1
Webcam	1
Computer/Laptop	1

4. Software Requirements-

Software	Purpose
Python 3.11	Main programming language
Open CV	Video Processing
MediaPipe	Hand Tracking
Pyserial	Serial Communication
Arduino IDE	Arduino simulation
Wokwi Simulator	Circuit Simulation
VS Code	Development Environment

5. Working Principle-

The webcam continuously captures video frames. MediaPipe detects hand landmarks from the video feed. The program compares fingertip positions with lower finger joints to determine whether each finger is open or closed.

Each finger state is converted into binary form:Example:

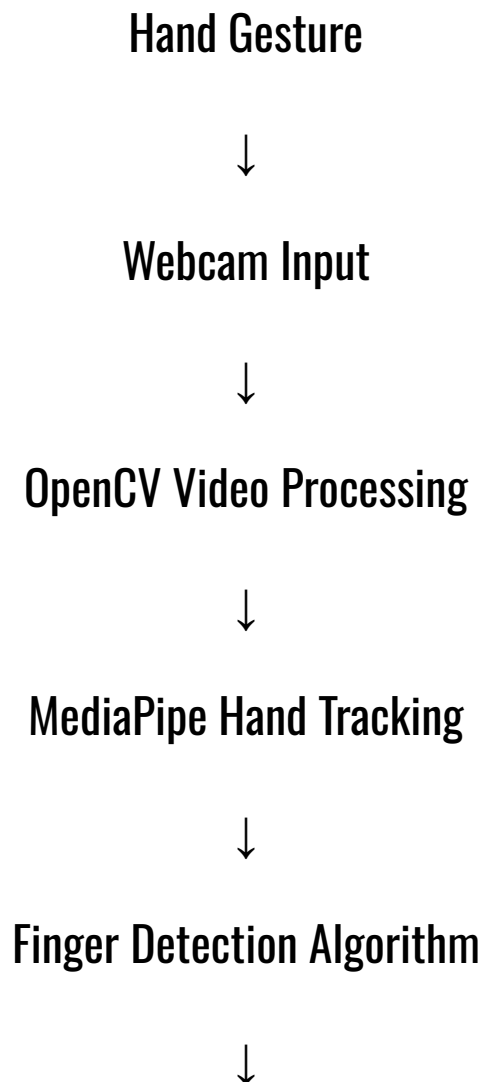
Finger	Binary Value
Thumb	1 or 0
Index	1 or 0
Middle	1 or 0
Ring	1 or 0
Pinky	1 or 0

- 11111 → All fingers open → All LEDs ON
- 00110 → Middle and Ring fingers open → Corresponding LEDs ON

The binary command is transmitted to Arduino through serial communication.

The Arduino receives the command and turns ON/OFF LEDs connected to digital pins.

6. System Architecture-



Binary Command Generation



Serial Communication



Arduino Uno



LED Control

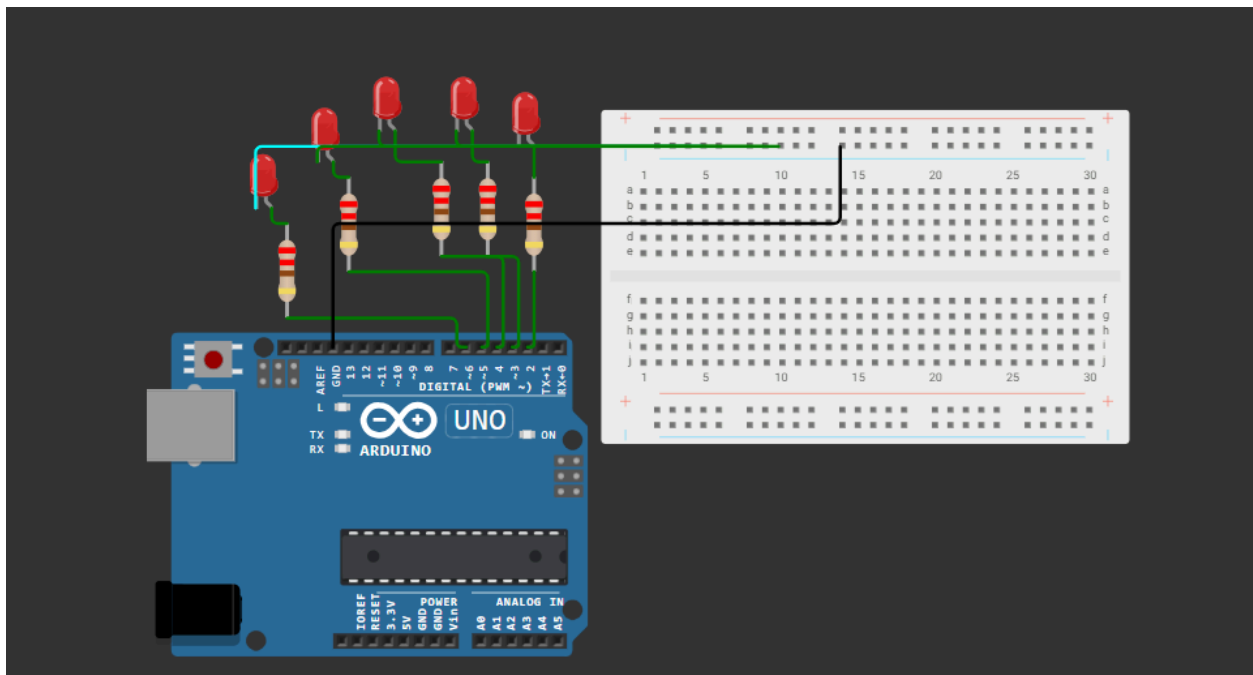
7. Circuit Connections-

LED	Arduino Pin
LED 1	Pin 2
LED 2	Pin 3

LED 3	Pin 4
LED 4	Pin 5
LED 5	Pin 6

All LEDs are connected in series with 220Ω resistors.

Cathodes are connected to GND.



8. Arduino Code Explanation-

The Arduino program:

1. Initializes serial communication.
2. Configures LED pins as OUTPUT.
3. Reads incoming binary commands.
4. Controls LEDs according to received data.

Example logic:

- If command bit is 1 → LED ON
- If command bit is 0 → LED OFF

9. Python Code Explanation-

The Python program performs:

9.1 Video Capture

OpenCV captures frames from the webcam.

9.2 Hand Detection

MediaPipe detects 21 hand landmarks.

9.3 Finger State Detection

The fingertip coordinates are compared with lower joints.

9.4 Command Generation

Finger states are converted into binary strings.

Example:

11111

00110

01000

9.5 Serial Communication

Commands are sent to Arduino using PySerial.

10. Algorithm-

Step-by-Step Algorithm

1. Start webcam.

2. Capture video frame.
3. Detect hand landmarks using MediaPipe.
4. Identify open and closed fingers.
5. Generate binary command.
6. Send command to Arduino.
7. Arduino controls LEDs.
8. Repeat continuously.
9. Stop when user presses Q.

11. Features of the project-

Real-time gesture recognition

Multi-finger detection

Left and right hand support

Real-time LED control

Serial communication interface

User-friendly graphical overlay

Low-cost implementation

12. Applications-

Smart home automation

Touchless control systems

Human-machine interaction

Robotics control

Industrial automation

Assistive technologies

Educational demonstrations

13. Advantages-

Contactless operation

Real-time response

Easy to implement

Low hardware cost

Scalable system design

Interactive user interface

14. Future Scope-

The project can be extended further using:

- ESP32 for wireless control
- Bluetooth or WiFi communication
- Home appliance automation
- Servo motor control
- Robotic arm operation
- AI-based gesture recognition
- Sign language interpretation
- IoT integration

15. Results-

The system successfully:

- Detected hand gestures in real time
- Identified individual finger states
- Generated binary commands accurately

- Communicated with Arduino successfully
- Controlled LEDs according to gestures

The project demonstrated stable and responsive gesture-based hardware control.

16. Conclusion-

This project successfully implemented a Gesture Controlled LED System using OpenCV, MediaPipe, Python, and Arduino Uno. The system accurately detected hand gestures and translated them into real-time LED control commands.

The project demonstrates the practical integration of computer vision and embedded systems for touchless interaction applications.

17. Project Images-